



Payments Clearinghouse



The Payments Clearinghouse is Part 1 of the [Livepeer: Weir](#) proposal.

Abstract

This document proposes a payment proxy mechanism for Livepeer, here called a clearinghouse. This clearinghouse would work on top of the existing Livepeer probabilistic micropayment (PM) system, signing PM tickets on behalf of gateways, while maintaining a running balance on behalf of those gateways. The clearinghouse could then settle the balance with its users in currencies such as USD, or draw down an account for grant recipients or other special-purpose entities. There would be no change to the core gateway-to-orchestrator payment flow, which would still happen with PM and settle via Eth. This approach also improves gateway security by separating out the payment signing key from untrusted media flows.

Background

How is it done today, and what are the limits?

The Livepeer probabilistic micropayments (PM) system has been a success since its introduction in 2019, securing millions of dollars of value transfer on the Livepeer network. While PM has been proven at scale, there are a few issues with the scheme that have slowed wider adoption of Livepeer:

Cryptocurrency

PM is built on top of Ethereum smart contracts, with Eth as the underlying currency of exchange.

Cryptocurrency still has a stigma among the wider public: individual developers who would otherwise be interested in Livepeer might find the crypto aspects unappealing.

Crypto is difficult for organizations to adopt from an operational perspective, because crypto does not easily fit into existing purchasing flows, and many organizations are not equipped to properly manage hot wallets.

PM

The design of PM is intricate and there are good reasons for its complexity, however this makes it harder for ordinary developers to integrate, even if these developers are crypto-friendly. There are many concepts to learn when setting up a Livepeer gateway with PM: the reserve and deposit amount, calculating expected value of tickets, the impact of gas, etc. At volume and as the network grows, the PM reserve balance may grow large, and this idle capital cannot easily be put towards other uses. Moreover, the friction in committing and withdrawing funds due to the unlock period discourages PM usage for one-off scenarios.

Gateway provider centralization

The factors of cryptocurrency and PM complexity, among others, have led the Livepeer network to centralize around a few large “gateway providers” that smooth over integration with the Livepeer network, including payments. Providers can then run customer billing in local currencies. However, these gateway providers are not structurally aligned with the network, and could just as easily take traffic away from the network as they could bring it. They may not always support the payment schemes that customers prefer. Customers also cannot control where a gateway provider is located or the features that they support.

Payment Signing Key in the Media Hot Path

Currently, the gateway must hold the payment signing key in the same process that manages untrusted media coming from users. This is a security risk, as an exploit stemming from processing untrusted input could result in the compromise of the payment signing key and lost gateway funds. Moreover, the design of PM leads gateway providers to share a common key for all gateway instances, which increases the blast radius of any compromise.

SPE Accountability and Capital Efficiency

Livepeer special-purpose entities often request LPT from a community Treasury for the purposes of funding network usage. This places an additional integration burden for SPEs that prefer to operate their own gateways: they must perform LPT → ETH conversion (including managing any tax liability from that), and locks up some of their funds via PM reserve / deposit, and manage Eth secret keys. Moreover, the community has limited visibility into how this granted LPT is spent and how much actually goes to network usage.

What's New

And how it will be successful

go-livepeer payment signer

The go-livepeer software can be extended with a new “payment signer” mode to sign PM tickets on demand, along with RPCs for media gateways to request signed PM tickets and any supporting signatures such as for [OrchestratorRequest](#). This separates the payment signing key from the media hot path, which addresses the security issue of keeping the payment signing key in the media hot path. This mode is also foundational for the operation of a clearinghouse, as signing PM tickets (and keeping balances topped up) the only function that a clearinghouse has on the network.

The payment signer mode only needs to implement basic signing for tickets and supporting RPCs such as [OrchestratorRequest](#). Media gateways can then send these requests and tickets to orchestrators in the same way they do now, without any changes to orchestrator job flows for on-chain jobs. This reduces rollout risk: remote signers are optional, and do not change the successful core payment protocol. This step also does not require additional bookkeeping on the gateway to keep track of signed tickets or customer balances. Clearinghouse implementations can add those important bookkeeping details at a later stage.

Local Gateway Support

Separating out payments from core media processing is an essential step in making Livepeer gateways easier to run locally. Gateways can request signed PM tickets from a payment service - all the operator needs to supply is an API key and / or a URL. This separates out the cryptocurrency aspects from the developer’s basic goal of integrating with the Livepeer network in order to perform AI or transcoding jobs. Running gateways locally helps the network cater to more use cases, such as those that are more latency-sensitive, or need keep more processing local for tighter integration, or developers that simply want to improve their security posture by separating out crypto hot wallets from untrusted media handling.

Grants Clearinghouse SPE

Operators that run payment signers - clearinghouses - can track and settle customer balances in a similar way that gateway providers do today, relying on the volume and frequency of PM ticket signing requests from a customer. These balances can be settled via USD, a local currency, or drawn down from a dedicated account for SPEs or grant recipients.

To further develop this approach, the community can fund a SPE to build and run its first clearinghouse to give credits for network usage. Rather than give new SPEs and other grant recipients LPT or Eth for network usage without further accountability, recipients could receive an account at the clearinghouse once their funding proposals have been accepted. Growth initiatives, such as hackathon projects or prizes, could also use this grants clearinghouse to give out usage credits.

The grants clearinghouse can then publish reports on network usage for transparency, so the community can understand the impact of various projects in terms of network fees. Recipients such as SPEs would not have to manage on-chain funds, and the clearinghouse can efficiently re-allocate unused funds towards other community initiatives, improving capital efficiency.

The grants clearinghouse can then act as a basis for other, more specialized clearinghouses, such as those that process fiat currencies.

Specialized Clearinghouses

The barrier to run a clearinghouse will be much lower than running a full-service gateway provider which must manage media, crypto and fiat payments. As such, we should see a diversity of clearinghouses appear, catering to specialized payment mechanisms. For example, a clearinghouse could settle with its customers in Bitcoin or Solana for cross-chain interoperability. Clearinghouses could settle with customers using [x402](#) or similar smart-contract backed protocols. Clearinghouses can be tied to the Livepeer platform in other ways, for example orchestrators could run their own clearinghouses to offer free or reduced price work to delegators. Local currency clearinghouses could make cross-border operations more tractable.

On-Chain Differentiation

To improve transparency, the community may opt to extend PM payments tickets to differentiate and track clearinghouse payments. Fields could be added with an ID for the clearinghouse customer, and an ID for the customer's end-user. Both fields should be optional and would be there to facilitate on-chain analysis. The clearinghouse should still be able to derive the same information via its own internal bookkeeping, but the community could verify the clearinghouse's own reports via on-chain data. This would work in conjunction with existing efforts to add AI job differentiation to PM tickets.

Structural Alignment

Clearinghouses are aligned with the network in a way that traditional full-service gateway providers are not, because the clearinghouse's reason for existence is to support payments on the Livepeer network. Gateway providers, however, can take their traffic in-house anytime, and bypass Livepeer, because they have no structural alignment with the network. Facilitating payments lowers the barriers to running local gateways, which in turn increases the number and variety of entry points to the network. Deliberate steps here can steer Livepeer away from becoming a consortium of incumbent and non-aligned gateway providers.

Who cares?

If this is successful, what difference will it make?

Payments clearinghouses have the potential to make Livepeer more accessible as a platform. This allows for developer interaction directly with the Livepeer network using their preferred currency, while masking the network's cryptocurrency underpinnings and the operational difficulties of managing a hot crypto wallet.

This would encourage gateways to run in more places, with a greater diversity of operators, utilizing more payment methods. Allowing for more network entry points means success for Livepeer can come from more directions: crypto-shy developers, ordinary organizations, and web3 developers that still don't really want to deal with crypto.

The clearinghouse concept offers another role on the network that is fully aligned with Livepeer: one that is focused on facilitating network payments. Operators would not have to take on the infrastructure, operations and business-development burden of becoming a full-service gateway provider. Clearinghouses also offer a space to experiment with novel payment and accounting mechanisms without requiring changes to core payment protocols.

The separated payment signer architecture also improves the overall security posture of gateway operators, which further increases trust and reputational standing of the Livepeer network and its tools.

Risks and Drawbacks

Clearinghouse centralization for a currency

There is a likelihood of centralization around clearinghouses for a specific currency, such as US dollars. With that, there is a risk that a clearinghouse for a certain currency being unavailable means that customers using that currency will not be able to run jobs on the Livepeer network. Careful design can mitigate the effects of short interruptions in clearinghouse availability, such as requesting a batch of tickets up front.

Moreover, since the RPC interface for clearinghouses to communicate with Livepeer media gateways would be well defined, it should be straightforward to switch clearinghouses if needed, such by changing a URL. This is unlike centralized provider APIs, which tend to be proprietary to the provider, and hence more difficult to swap out.

A clearinghouse should also be easier to operate than a full-service gateway provider who must also deal with performance-sensitive media workflows. Hence, the community may deem it wise to ensure there are multiple independent clearinghouses for key currencies.

Under an expansive interpretation of the clearinghouse's role, the clearinghouse could refuse to sign payments for certain orchestrators it deems unsuitable for the network.



Disjoint relationship between network usage and billing

The relationship with the clearinghouse - the billing provider for the network - does not necessarily relate to a customer's operational experience on the network. This could be a problem for customers that expect a more traditional and fully integrated service and support system.

For example, payments are still mostly irrevocable due to Livepeer's Ethereum underpinnings. Dispute resolution is less straightforward since the payment is decoupled from the actual work being done. Clearinghouses cannot easily credit or refund customers for orchestrator performance issues, catastrophic SDK bugs, or customer mistakes without eating the pass-through cost to the orchestrator. This is an inherited problem from crypto payments which the clearinghouse architecture cannot solve on its own. For significant, widespread incidents that might require SLA-like credits, the community could (pre-)fund a sort of reinsurance pool for clearinghouses to tap into.

Self Dealing Orchestrators

The clearinghouse does not solve the issue of self-dealing orchestrators. With free services such as Daydream, orchestrators could select themselves and get paid to do nothing. Hence, gateway operators will still have to take an active role in quality monitoring.

Clearinghouses could publish transparency reports, either ecosystem-wide or per customer app, to help identify suspicious patterns of behavior. They could also more active role in quality monitoring, fraud detection, and recommending which orchestrators to use.

How long will it take?

- Go-live/peer implementation for the payment signer mode, along with testing, required documentation updates and best-practices messaging: **1 month**
- Grants clearinghouse: **1 month** for the SPE proposal and a design that includes user account management, balance tracking and a system to generate reports for the community. **3 months** to build and launch, plus ongoing maintenance under a Grants Clearinghouse SPE.
- USD billing service: **3 months** to launch using the grants service as a base. Operations should be run by a legal entity rather than a SPE. This could involve seeding a company via Treasury proposal, or partnering with an existing entity such as Inc.